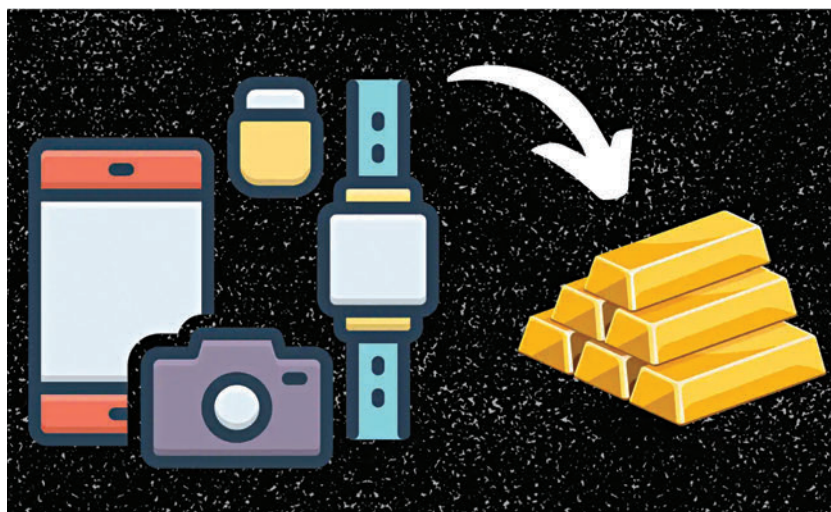




Finding value in e-waste

How can recycling gold from e-waste transform sustainable chemistry?

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E-waste is often considered an environmental hazard, but recent advancements in recycling technology have revealed its potential as a goldmine – quite literally. Globally, we generate an estimated 50 million tons of e-waste annually, a figure expected to double by 2050. This rapidly growing waste stream contains valuable resources like gold, silver, and copper, with over \$57 billion worth of recoverable materials discarded each year. In their paper, titled – “Recycling e-waste into gold-loaded covalent organic framework catalysts for terminal alkyne carboxylation” – researchers from Cornell University published their findings about

advanced technologies to tap into this resource, turning discarded electronics into catalysts for critical chemical processes.

Every year, we generate a staggering 50 million tons of e-waste globally, presenting not just an environmental challenge but also a golden opportunity for resource recovery. Scientists have been pushing the boundaries of material science to extract gold from discarded electronics, turning this waste into catalysts for essential chemical processes. At the forefront of this transformation are covalent organic frameworks (COFs), cutting-edge materials engineered to extract and repurpose gold ions with incredible precision.

Mining gold from electronics

Gold's allure isn't just aesthetic; it's critical in electronics for its exceptional conductivity and resistance to corrosion. However, mining it traditionally is an environmental nightmare, often involving cyanide-based leaching and large-scale deforestation. Enter “urban mining,” a sustainable approach that focuses on recovering precious metals from e-waste. This concept is quickly gaining traction as industries seek greener alternatives to conventional mining. Think about this: the average smartphone contains around 0.034 grams of gold. While this seems trivial, multiply it by billions of devices, and the numbers add up. Current recycling methods recover about 25% of the world's gold, but they struggle with efficiency and selectivity when dealing with complex metal mixtures. This is where advanced materials like COFs come into play, offering a game-changing solution for gold recovery.

Why COFs are a game-changer?

COFs, or covalent organic frameworks, are essentially molecular sieves on steroids. These materials are highly porous, lightweight, and chemically tunable, making them perfect for capturing specific metal ions. Two COFs, in particular, have stolen the spotlight: TTF-COF and TPE-COF. Designed using tetrathiafulvalene (TTF) and tetraphenylethylene (TPE), these frameworks are engineered for maximum efficiency in gold adsorption.

Here's the kicker – TTF-COF's sulfur-rich structure interacts with gold ions through a soft acid-base interaction. This allows it to adsorb gold with jaw-dropping efficiency – 2440 mg of gold per gram of COF. To put that into perspective, it can capture over 99% of gold from a solution, even when competing metals like nickel and copper are